SMALL PRIMES FORCE $C_{1/(m+1)} \geq 2^m$: THE FORMAL "OPTIMAL" VARIANT OF ERDŐS PROBLEM #1180 IS FALSE

MATH0

Abstract. For $\varepsilon > 0$ let C_ε be the least C such that, for every prime p, every residue modulo p is a sum of at most C inverses n^{-1} mod p with $1 \leq n \leq p^\varepsilon$. The formal-conjectures statement `Erdos1180.erdos_1180.variants.optimal` asserts that for every $\delta > 0$ one has $C_\varepsilon \leq \varepsilon^{-1-\delta}$ for all small $\varepsilon > 0$; we prove it false by showing $C_{1/(m+1)} \geq 2^m$ for every $m \geq 2$, since for $\varepsilon = 1/(m+1)$ and a Bertrand prime $2^m < p < 2^{m+1}$ the only admissible denominator is 1 (as C_ε is a minimum, this also uses a proof, along Glibichuk's lines, that a finite bound exists). This is a misformalization report, not progress on the research problem: the intended question, whether $\varepsilon^{-1-o(1)}$ summands suffice for all sufficiently large primes, remains open, the small-prime obstruction is recorded upstream, and a Lean disproof of a verbatim copy of the statement already exists (linrock/math-proofs). We give an independent proof, verified in Lean 4 against the upstream formal-conjectures statement at commit `c77fdd4`, and propose a corrected statement.

1. Introduction

Erdős and Graham [2, p. 103] asked the following question, listed as Erdős Problem #1180 [1]: given $\varepsilon > 0$, is there a constant C_ε such that, for every prime p, every residue modulo p is a sum of at most C_ε elements of $\{n^{-1} : 1 \leq n \leq p^\varepsilon\}$, where n^{-1} is the inverse of n modulo p?

Known results. Shparlinski [12] answered the question affirmatively, even with pairwise distinct denominators. The problem page [1] records the quantitative form $C_\varepsilon \ll \varepsilon^{-3}$ of Shparlinski's result, and Glibichuk's improvement to $\ll \varepsilon^{-2}$ [6]: for every sufficiently large prime p every residue is a sum of $8(\lfloor 1/\varepsilon + 1/2 \rfloor + 1)^2$ such inverses. As in [6], and as the upstream formal statements record for both bounds, these quantitative bounds concern sufficiently large primes $p \geq P(\varepsilon)$. The page also records a result of Croot with $(\log p)^{3+o(1)}$ summands, notes that $\gg \varepsilon^{-1}$ summands are trivially necessary, and remarks that "it may be that $C_\varepsilon \leq \varepsilon^{-1-o(1)}$ is possible". The main question is thus settled, and the last remark is the open part of the problem.

The formal statements and prior work. The formal-conjectures repository [3] formalizes the problem in `FormalConjectures/ErdosProblems/1180.lean`, with C_ε defined as the least bound valid for *all* primes. The main statement `erdos_1180` and the variants `shparlinski`, `glibichuk` and `lower_bound` are tagged *solved*, with linked Lean proofs in plby/lean-proofs [11] and linrock/math-proofs [7]; the variant `erdos_1180.variants.optimal`, which asks whether $C_\varepsilon \leq \varepsilon^{-1-\delta}$ for every $\delta > 0$ and all small ε, is tagged *research open*. The obstruction used here is known upstream: commit `377a7b38` (pull request #6486) [4] restated `shparlinski` and `glibichuk` for large primes only, because no bound on the all-primes constant can be polynomial in ε^{-1}, as 1 is the only admissible denominator when $p^\varepsilon < 2$, and its message notes that "the audit also flags `variants.optimal` as questionable for the same reason (it uses the all-primes `C`)", but leaves that statement unchanged because the problem page reuses C_ε there without a prime threshold.

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A Lean disproof already exists: linrock/math-proofs [7], at commit `ba96303b`, proves ¬`Erdos1180OptimalConjecture`, a verbatim copy of the upstream statement and its definitions, as `Erdos1180.Palomar.not_erdos_1180_variants_optimal`, together with $2^{1/\varepsilon} \le 4C_\varepsilon$. This work is registered as Palomar entry PALOMAR-2026-10-12-000005 [10], cited in the upstream file; pull request #7219 [5] linked it for the three other variants, while `variants.optimal` carries no link at commit `c77fdd4`. Our refutation is therefore not the first formal one; it is independent, and it is checked against the upstream declaration itself rather than a copy. The part of our Lean file that proves `erdos_1180` (Glibichuk's argument, Sections 3.1–3.3) is adapted, under its Apache-2.0 licence, from plby/lean-proofs [11], with lists replaced by multisets. We classify the result as a *misformalization report*: the formal statement fails because it quantifies over all primes, including those below $2^{1/\varepsilon}$, while the open question concerns large primes. Section 4 proposes a corrected statement.

2. Statement

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, and powers x^y with real exponent are real powers. Let $\varepsilon \in \mathbb{R}$, let p be a prime and $a \in \mathbb{F}_p$. An *ε-representation of a* is a finite multiset s of natural numbers such that every $n \in s$ satisfies $1 \le n \le p^\varepsilon$ and $\gcd(n, p) = 1$ (we call such n *admissible*), and $\sum_{n \in s} n^{-1} = a$ in $\mathbb{F}_p$, counted with multiplicity. Let $S_\varepsilon \subseteq \mathbb{N}$ be the set of C such that every $a \in \mathbb{F}_p$ has an ε-representation of size at most C, for every prime p, and put $C_\varepsilon = \min S_\varepsilon$ if $S_\varepsilon \neq \emptyset$ and $C_\varepsilon = 0$ otherwise. The upstream statement asserts:

$$\forall \delta > 0 \;\; \exists \eta > 0 \;\; \forall \varepsilon \in (0, \eta) : \quad C_\varepsilon \le \varepsilon^{-(1+\delta)}. \tag{1}$$

The definitions and the statement, copied verbatim from `FormalConjectures/ErdosProblems/1180.lean` at commit `c77fdd4` (inside `namespace Erdos1180`, with `open Filter`; docstrings of the two definitions omitted), read:

```
def Represents (ε : ℝ) (p : ℕ) (a : ZMod p) (s : Multiset ℕ) : Prop :=
  (∀ n ∈ s, 1 ≤ n ∧ (n : ℝ) ≤ (p : ℝ) ^ ε ∧ n.Coprime p) ∧
    (s.map fun n : ℕ ↦ (n : ZMod p)⁻¹).sum = a

noncomputable def C (ε : ℝ) : ℕ :=
  sInf {C | ∀ p : ℕ, p.Prime → ∀ a : ZMod p, ∃ s : Multiset ℕ, s.card ≤ C ∧ Represents ε p a
    s}

/-- Is $C_\epsilon\leq \epsilon^{-1-o(1)}$? -/
@[category research open, question, AMS 11]
theorem erdos_1180.variants.optimal : ∀ δ : ℝ, 0 < δ →
    ∀ᶠ ε : ℝ in nhdsWithin 0 (Set.Ioi 0), (C ε : ℝ) ≤ ε⁻¹ ^ (1 + δ) := by
  sorry
```

Since `sInf` $\emptyset = 0$ in $\mathbb{N}$, the filter `nhdsWithin 0 (Set.Ioi 0)` consists of the sets containing some $(0, \eta)$, and $(\varepsilon^{-1})^{1+\delta} = \varepsilon^{-(1+\delta)}$ for $\varepsilon > 0$, the Lean statement is exactly (1).

Theorem 1. *Statement* (1) *is false. In fact it fails for $\delta = 1$: $C_{1/(m+1)} > (m+1)^2$ for every $m \ge 6$.*

In Lean, Theorem 1 is `Math0.Erdos1180.not_erdos_1180_variants_optimal`, whose type is ¬ `(type_of% @_root_.Erdos1180.erdos_1180.variants.optimal)`, the negation of the upstream statement. The proof rests on two facts; the first is needed because if S_ε were empty, then $C_\varepsilon = 0$ and the inequality in (1) would hold trivially.

Theorem 2 (after Glibichuk [6]; upstream `erdos_1180`)**.** *Let $\varepsilon > 0$ and let k be an integer with $k > 2/\varepsilon + 1$. There is P_0 such that for every prime $p \geq P_0$ every $a \in \mathbb{F}_p$ has an ε-representation of size exactly $8k^2$. Consequently $\max(8k^2, P_0) \in S_\varepsilon$; in particular $S_\varepsilon \neq \emptyset$.*

Proposition 3. *For every integer $m \geq 2$, $C_{1/(m+1)} \geq 2^m$.*

3. Proof

3.1. **Glibichuk's covering lemma.** Let p be prime and $X, Y \subseteq \mathbb{F}_p$ with $N = |X||Y|$. For $\xi \in \mathbb{F}_p$ let $A_\xi = \{x + \xi y : x \in X,\, y \in Y\}$ and let E_ξ be the number of pairs $(u, v) \in (X \times Y)^2$ with $u_1 + \xi u_2 = v_1 + \xi v_2$. Write $XY = \{xy : x \in X, y \in Y\}$ and hXY for the h-fold sumset $XY + \cdots + XY$. Call Y *antisymmetric* if $-y \notin Y$ for every $y \in Y$.

Lemma 4. (a) $N^2 \leq |A_\xi|\, E_\xi$ *for every* ξ. (b) $\sum_{\xi \neq 0} E_\xi \leq (p-1)N + N^2$*; hence some* $\xi \neq 0$ *satisfies* $E_\xi \leq N + N^2/(p-1)$. (c) $E_{-\xi} = E_\xi$.

Proof. (a) For $z \in A_\xi$ let $r(z)$ be the number of $u \in X \times Y$ with $u_1 + \xi u_2 = z$. Then $\sum_z r(z) = N$ and $\sum_z r(z)^2 = E_\xi$, and Cauchy–Schwarz gives $N^2 \leq |A_\xi| \sum_z r(z)^2$. (b) The sum counts triples (ξ, u, v) with $\xi \neq 0$ and $u_1 + \xi u_2 = v_1 + \xi v_2$. Triples with $u = v$ number at most $(p-1)N$. If $u \neq v$ and some ξ solves the equation, then $u_2 \neq v_2$ (otherwise the equation forces $u_1 = v_1$), and two solutions ξ, η give $(\xi - \eta)(u_2 - v_2) = 0$, so $\xi = \eta$; hence these triples number at most N^2. The second claim follows by averaging over the $p - 1$ values $\xi \neq 0$. (c) The involution $((x_1, y_1), (x_2, y_2)) \mapsto ((x_1, y_2), (x_2, y_1))$ of $(X \times Y)^2$ maps solutions for ξ to solutions for $-\xi$, since $x_1 + \xi y_1 = x_2 + \xi y_2$ is equivalent to $x_1 - \xi y_2 = x_2 - \xi y_1$. □

Lemma 5. *Let $p > 2$ be prime, $N > p$ and $E_\xi \leq N + N^2/(p-1)$. Then $2|A_\xi| > p$.*

Proof. Suppose $2|A_\xi| \leq p$. As p is odd, $2|A_\xi| \leq p - 1$. By Lemma 4(a), $2N^2 \leq 2|A_\xi| E_\xi \leq (p-1)E_\xi \leq (p-1)N + N^2$, so $N \leq p - 1$, a contradiction. □

Proposition 6 (Glibichuk)**.** *Let $p > 2$ be prime and $X, Y \subseteq \mathbb{F}_p$ with $|X||Y| > p$ and Y antisymmetric. Then $8XY = \mathbb{F}_p$.*

Proof. By Lemmas 4 and 5 there is $\xi \neq 0$ with $2|A_\xi| > p$ and $2|A_{-\xi}| > p$. A set $A \subseteq \mathbb{F}_p$ with $2|A| > p$ meets $-A$, since otherwise $|A \cup (-A)| = 2|A| > p$. So $x_1 + \xi y_1 = -(x_2 + \xi y_2)$ for some $x_i \in X$, $y_i \in Y$, that is, $\xi(y_1 + y_2) = -(x_1 + x_2)$, and $c = y_1 + y_2 \neq 0$ because Y is antisymmetric. For $x_3 \in X$, $y_3 \in Y$,

$$c\,(x_3 - \xi y_3) = x_3 y_1 + x_3 y_2 + x_1 y_3 + x_2 y_3 \in 4XY.$$

Multiplication by c is injective, so $|4XY| \geq |A_{-\xi}| > p/2$, i.e. $2|4XY| \geq p + 1$. By the Cauchy–Davenport theorem, $|4XY + 4XY| \geq \min(p, 2|4XY| - 1) = p$, so $8XY = \mathbb{F}_p$. □

3.2. **Sums of reciprocals of primes in blocks.** Let p be prime, $k \geq 1$, $L \geq 0$, $H < p$, and let $q_{i,j}$ $(0 \leq i < k,\, 0 \leq j < L)$ be pairwise distinct primes with $q_{i,j} \leq H$ and

$$2kH^{2k-1} < p. \tag{2}$$

For $u\colon \{0, \ldots, k-1\} \to \{0, \ldots, L-1\}$ put $r(u) = \sum_i q_{i,u(i)}^{-1} \in \mathbb{F}_p$, and let $B = \{r(u)\}$.

Lemma 7. *The map $u \mapsto r(u)$ is injective, so $|B| = L^k$, and B is antisymmetric.*

Proof. Let $D(u) = \prod_i q_{i,u(i)}$ and $M(u) = \sum_i \prod_{j \neq i} q_{j,u(j)}$, both positive integers. Since $0 < q_{i,j} < p$, each $q_{i,j}$ is invertible in $\mathbb{F}_p$ and $D(u)r(u) = M(u)$ in $\mathbb{F}_p$. Moreover $D(u) \leq H^k$ and $M(u) \leq kH^{k-1}$, so $0 < M(u)D(v) \leq kH^{2k-1} < p/2$ for all u, v.

Injectivity. If $r(u) = r(v)$ then $M(u)D(v) = D(u)D(v)r(u) = M(v)D(u)$ in $\mathbb{F}_p$; both sides are integers in $[0, p)$, so they are equal. Suppose $u \neq v$, choose i with $u(i) \neq v(i)$ and let $q = q_{i,u(i)}$.

Then $q \mid D(u)$, so q divides $M(v)D(u) = M(u)D(v)$. But $q \nmid D(v)$, since a prime dividing $D(v)$ equals some $q_{j,v(j)}$, and distinctness would give $(i, u(i)) = (j, v(j))$. Also $q \nmid M(u)$: every summand of $M(u)$ except the i-th contains the factor q, and the i-th is a product of primes $q_{j,u(j)} \neq q$. This contradicts the primality of q.

Antisymmetry. If $r(u) = -r(v)$ then p divides $M(u)D(v) + M(v)D(u)$, an integer in $(0, p)$; impossible. □

Lemma 8. *Assume moreover $H^2 \leq p^\varepsilon$. Then every element of $8BB$ has an ε-representation of size exactly $8k^2$.*

Proof. For u, v we have $r(u)r(v) = \sum_{i,j}(q_{i,u(i)}q_{j,v(j)})^{-1}$, a sum of k^2 inverses. Each denominator $n = q_{i,u(i)}q_{j,v(j)}$ satisfies $1 \leq n \leq H^2 \leq p^\varepsilon$, and $\gcd(n, p) = 1$ because both factors are positive and less than p. So every element of BB has an ε-representation of size exactly k^2. The sum of the multisets of two representations represents the sum of the two residues, so every element of $8BB$ has one of size exactly $8k^2$. □

3.3. **Parameters and proof of Theorem 2.**

Lemma 9. *Let $\varepsilon > 0$ and let k be an integer with $k > 2/\varepsilon + 1$. Put $\delta = 2/(4k-1)$, $\gamma = (\delta + \frac{1}{2k})/2$ and $t = (\gamma/\delta + 1)/2$. Then $0 < \gamma < \delta < 1$, $(2k-1)\delta < 1$, $2\delta \leq \varepsilon$, $2k\gamma > 1$, $0 < t < 1$ and $\gamma < \delta t$.*

Proof. $\frac{1}{2k} < \delta$ because $4k - 1 < 4k$, so $\frac{1}{2k} < \gamma < \delta$ and $2k\gamma > 1$. Next $\delta < 1$ as $k > 1$, and $(2k-1)\delta = \frac{4k-2}{4k-1} < 1$. From $k\varepsilon > 2 + \varepsilon$ we get $\varepsilon(4k-1) > 8 + 3\varepsilon > 4$, i.e. $2\delta \leq \varepsilon$. Finally $\gamma/\delta \in (0, 1)$ gives $t \in (0, 1)$, and $\delta t = (\gamma + \delta)/2 > \gamma$. □

Lemma 10. *With the parameters of Lemma 9, for all sufficiently large n, writing $H = \lfloor n^\delta \rfloor$ and $L = \lfloor n^\gamma \rfloor$: (G1) $n > 2$; (G2) $H < n$; (G3) $2kH^{2k-1} < n$; (G4) $H^2 \leq n^\varepsilon$; (G5) $n < L^{2k}$; (G6) $kL \leq \pi(H)$.*

Proof. (G2) holds since $\delta < 1$. (G3): $2kH^{2k-1} \leq 2k\, n^{(2k-1)\delta} < n$ for large n, as $(2k-1)\delta < 1$. (G4): $H^2 \leq n^{2\delta} \leq n^\varepsilon$ for $n \geq 1$. (G5): once $n^\gamma \geq 1$ we have $L > n^\gamma/2$, so $L^{2k} > 2^{-2k} n^{2k\gamma} > n$ for large n, as $2k\gamma > 1$. (G6): we first show $\pi(x) \geq x^t$ for all large integers x. Chebyshev's bound $\pi(x) \geq (x \log 2 - \log(x+1))/\log x$ holds for all integers $x \geq 2$; for $x \geq 6$ we have $(x+1)^2 \leq 2^x$, so $\log(x+1) \leq \frac{x \log 2}{2}$, and for large x also $\log x \leq \frac{\log 2}{2} x^{1-t}$, whence $x^t \log x \leq \frac{x \log 2}{2} \leq x \log 2 - \log(x+1)$. Since $H \to \infty$, for large n we get $\pi(H) \geq H^t > (n^\delta/2)^t = 2^{-t} n^{\delta t} > k\, n^\gamma \geq kL$, using $H > n^\delta/2$ and $\delta t > \gamma$. □

Proof of Theorem 2. Take P_0 such that every $n \geq P_0$ satisfies (G1)–(G6). Let $p \geq P_0$ be prime and $a \in \mathbb{F}_p$. By (G6) there are kL distinct primes $\leq H$; label them $q_{i,j}$ $(i < k,\ j < L)$. By (G2) and (G3) the hypotheses of Section 3.2 hold, and Lemma 7 and (G5) give an antisymmetric B with $|B|^2 = L^{2k} > p$. Proposition 6 with $X = Y = B$ (using (G1)) gives $8BB = \mathbb{F}_p$, and Lemma 8 (using (G4)) gives an ε-representation of a of size $8k^2$. For a prime $p < P_0$ and $a \in \mathbb{F}_p$, let $0 \leq a' < p$ be the integer representing a; the multiset consisting of a' copies of 1 is an ε-representation of a (note $1 \leq p^\varepsilon$), of size $a' < P_0$. Hence $\max(8k^2, P_0) \in S_\varepsilon$. □

3.4. **The lower bound and proof of Theorem 1.**

Lemma 11. *Let $k \geq 1$ and let p be a prime with $p < 2^k$. Then every $(1/k)$-representation s of -1 has $|s| \geq p - 1$.*

Proof. If $n \in s$ had $n \geq 2$, then $2 \leq p^{1/k}$, and raising to the k-th power gives $2^k \leq p$, a contradiction. So s consists of $|s|$ copies of 1, and $\sum_{n \in s} n^{-1} = |s| = -1$ in $\mathbb{F}_p$. Thus $p \mid |s| + 1$, and $|s| + 1 \geq p$. □

Proof of Proposition 3. Let $m \geq 2$ and $\varepsilon = 1/(m+1)$. By Bertrand's postulate there is a prime p with $2^m < p \leq 2^{m+1}$; in fact $p < 2^{m+1}$, since $p > 2^m \geq 4$ is odd while 2^{m+1} is even. By Theorem 2, $S_\varepsilon \neq \emptyset$, so $C_\varepsilon = \min S_\varepsilon \in S_\varepsilon$, and $-1 \in \mathbb{F}_p$ has an ε-representation s with $|s| \leq C_\varepsilon$. By Lemma 11 with $k = m+1$, $C_\varepsilon \geq |s| \geq p - 1 \geq 2^m$. □

Proof of Theorem 1. Suppose (1) holds and apply it with $\delta = 1$. Since $\varepsilon_m = 1/(m+1)$ tends to 0 through positive values, there is M with $C_{\varepsilon_m} \leq \varepsilon_m^{-2} = (m+1)^2$ for all $m \geq M$. Take $m = M + 10$. Proposition 3 gives $2^m \leq C_{\varepsilon_m} \leq (m+1)^2$. But $(m+1)^2 < 2^m$ for $m \geq 6$, by induction: $49 < 64$, and $(m+2)^2 \leq 2(m+1)^2 < 2 \cdot 2^m$. Contradiction. □

The Lean proof carries out the argument of Proposition 3 inline for $m = M + 10$, using Mathlib's `Nat.exists_prime_lt_and_le_two_mul` (Bertrand) and `Nat.sInf_mem`.

4. The defect and a corrected statement

The lower bound of Proposition 3 comes only from primes $p < 2^{1/\varepsilon}$, for which 1 is the only admissible denominator (Theorem 2 only ensures that C_ε is a genuine minimum); any bound valid for all primes is at least $p-1$ for each such p, so C_ε is exponential in $1/\varepsilon$ whatever happens for large primes. The bounds of Shparlinski and Glibichuk concern the least bound valid for all primes $p \geq P(\varepsilon)$, for which $\gg \varepsilon^{-1}$ summands are also trivially necessary, and we read the remark "it may be that $C_\varepsilon \leq \varepsilon^{-1-o(1)}$" as referring to that quantity, since for the all-primes constant it fails trivially. Theorem 1 says nothing about the large-prime quantity, and whether it is $\varepsilon^{-1-o(1)}$ remains open.

Following the form that [4] gave to the `shparlinski` and `glibichuk` variants, we propose:

```
/-- Is $C_\epsilon\leq \epsilon^{-1-o(1)}$, for all sufficiently large primes? -/
@[category research open, question, AMS 11]
theorem erdos_1180.variants.optimal : ∀ δ : ℝ, 0 < δ →
    ∀ᶠ ε : ℝ in nhdsWithin 0 (Set.Ioi 0), ∃ P : ℕ, ∀ p : ℕ, P ≤ p → p.Prime → ∀ a : ZMod p,
      ∃ s : Multiset ℕ, (s.card : ℝ) ≤ ε⁻¹ ^ (1 + δ) ∧ Represents ε p a s := by
  sorry
```

We checked that this elaborates against the pinned versions of Section 5. Alternatively, one can bound a large-prime constant in place of `C`, as linrock/math-proofs does with `CAsymptotic` [7].

5. Machine verification

Files and declarations. The proof is the file `lean/Math0/ErdosProblems/Erdos1180_erdos_1180_variants_optimal.lean` (1073 lines) of the math0 repository (`https://github.com/Foundation-0/math0`). It imports only the upstream file `FormalConjectures/ErdosProblems/1180.lean` and the audit commands `lean/Math0/Audit.lean`, and contains no `sorry`, `native_decide`, `set_option` or new axioms. The main declarations, in the namespace `Math0.Erdos1180`, are: `exists_good_slope`, `affEnergy_neg` (Lemma 4); `card_affImg_large` (Lemma 5); `glibichuk` (Proposition 6, using Mathlib's `ZMod.cauchy_davenport`); `recSum_injective`, `card_recSet`, `recSet_antisymm` (Lemma 7); `sum8_exactRep` (Lemma 8); `choose_params` (Lemma 9); `ev_good`, `ev_primeCounting_ge` (Lemma 10, using Mathlib's `Chebyshev.pi_ge`); `exists_bound` (Theorem 2); `card_ge_of_represents_neg_one` (Lemma 11); `not_erdos_1180_variants_optimal` (Theorem 1). The statement of `exists_bound` is that of upstream `erdos_1180`; only the final theorem is audited against an upstream declaration, by the last line of the file,

```
#math0_refutes Math0.Erdos1180.not_erdos_1180_variants_optimal Erdos1180.erdos_1180.variants.optimal
```

Audit. The command `#math0_refutes` (defined in `lean/Math0/Audit.lean`) fails unless the two theorems have the same universe parameters, the type of ours is syntactically identical (up to metadata, with no unfolding or definitional-equality check) to $\neg T$ for T the type of `Erdos1180.erdos_1180.variants.optimal`, and ours depends on no axioms other than `propext`, `Classical.choice` and `Quot.sound`; this rejects `sorryAx`, the axioms behind `native_decide` and custom axioms. On this file it reports

```
math0 audit: 'Math0.Erdos1180.not_erdos_1180_variants_optimal' depends on axioms [Classical.choice, Quot.
    sound, propext]
math0 statement check: 'Math0.Erdos1180.not_erdos_1180_variants_optimal' refutes 'Erdos1180.erdos_1180.
    variants.optimal'
```

Toolchain and reproduction. We use Lean 4 [9] `v4.33.1`, Mathlib [8] `v4.33.1` and formal-conjectures [3] at commit `c77fdd446d23539302aa73bcfae4a9fa962b8382`, all pinned in `lean/`. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib cache, builds all math0 results with `lake build` (re-running every audit), and checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified. The single file can be checked in about a minute with `lake env lean` from `lean/`.

References

[1] T. F. Bloom, *Erdős Problems*, Problem #1180, `https://www.erdosproblems.com/1180`, accessed October 2026.

[2] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L'Enseignement Mathématique **28**, Université de Genève, 1980.

[3] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An open and evolving benchmark for verified discovery in mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[4] formal-conjectures, commit `377a7b38` (pull request #6486), *fix(ErdosProblems/1180): bound summands for large primes only*, 22 September 2026, `https://github.com/google-deepmind/formal-conjectures/pull/6486`.

[5] formal-conjectures, commit `da0010d4` (pull request #7219), *feat(ErdosProblems/1180): link formal proofs for solved variants*, 13 October 2026, `https://github.com/google-deepmind/formal-conjectures/pull/7219`.

[6] A. A. Glibichuk, *Combinatorial properties of sets of residues modulo a prime and the Erdős–Graham problem*, Mat. Zametki **79** (2006), no. 3, 384–395; English transl. in Math. Notes **79** (2006), no. 3, 356–365.

[7] linrock, *math-proofs*, directory `erdos-1180/` at commit `ba96303b34a95ac458958c1eb0bbdee54733484a`, `https://github.com/linrock/math-proofs`.

[8] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

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[10] Palomar registry, entry PALOMAR-2026-10-12-000005, version 1, `https://palomar-registry.org/entry.html?id=PALOMAR-2026-10-12-000005&version=1`.

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